

Power: 10-20V input / 5V / separate 3V3

File: 01-power.kicad_sch

Teensy 4.1 / expansion connectors

File: 02-mcu.kicad_sch

CrowPanel Advance J10: 5V power and translated UART

File: 03-screen.kicad_sch

External optocoupler and GPS input buffers

File: 04-inputs.kicad_sch

Power: 10–20V input / 5V / separate 3V3

A

B

C

D

E

F

A

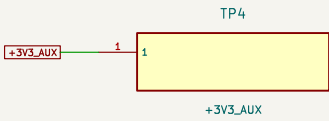
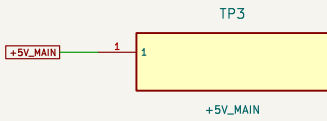
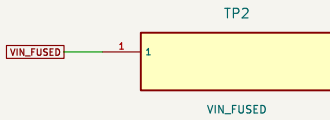
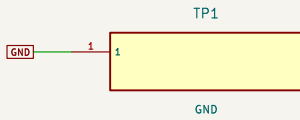
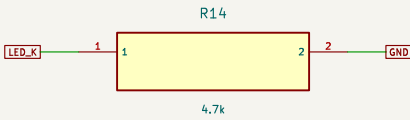
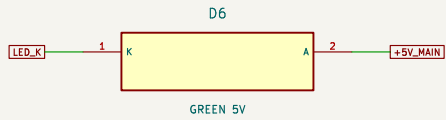
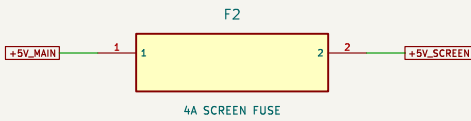
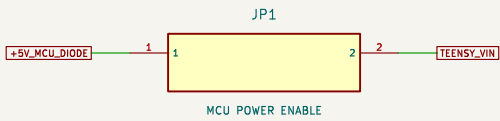
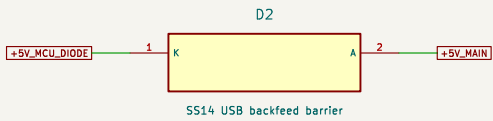
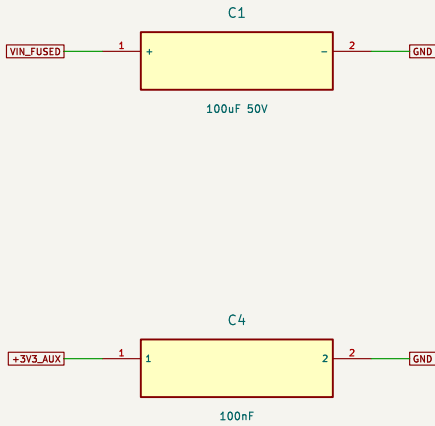
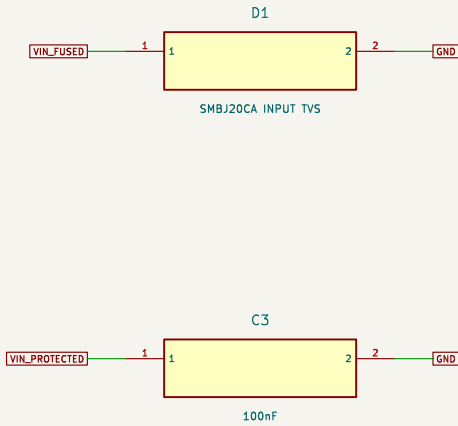
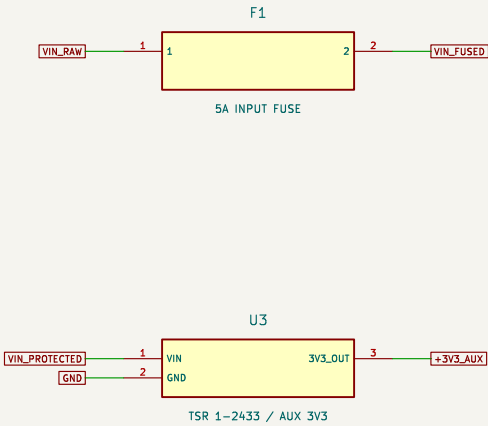
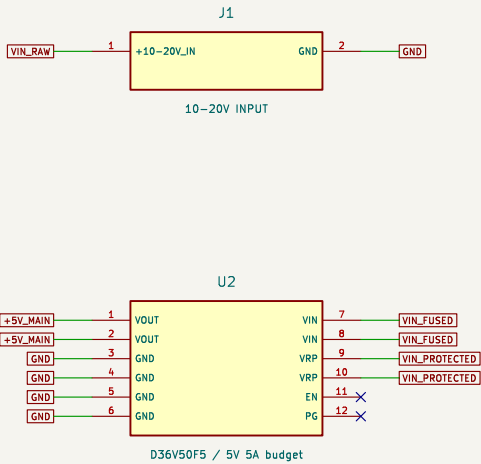
B

C

D

E

F



UNVALIDATED PROTOTYPE. Regulator module provides 5V: PCB total 5V budget 5A, screen branch fused 4A.

SMBJ20CA + fuse is NOT ISO 7637 / ISO 16750 load–dump qualification. Add harness fuse near battery.

U3 takes VIN_PROTECTED from Peliolu VRP. AUX 3V3 and Teensy 3V3 are DIFFERENT rails.

JP1 is Initial–test Isolation only. CUT the Teensy VUSB–VIN bridge before dual USB/carrier power.

Sheet: /Power: 10–20V input / 5V / separate 3V3/
File: 01–power.kicad_sch

Title:

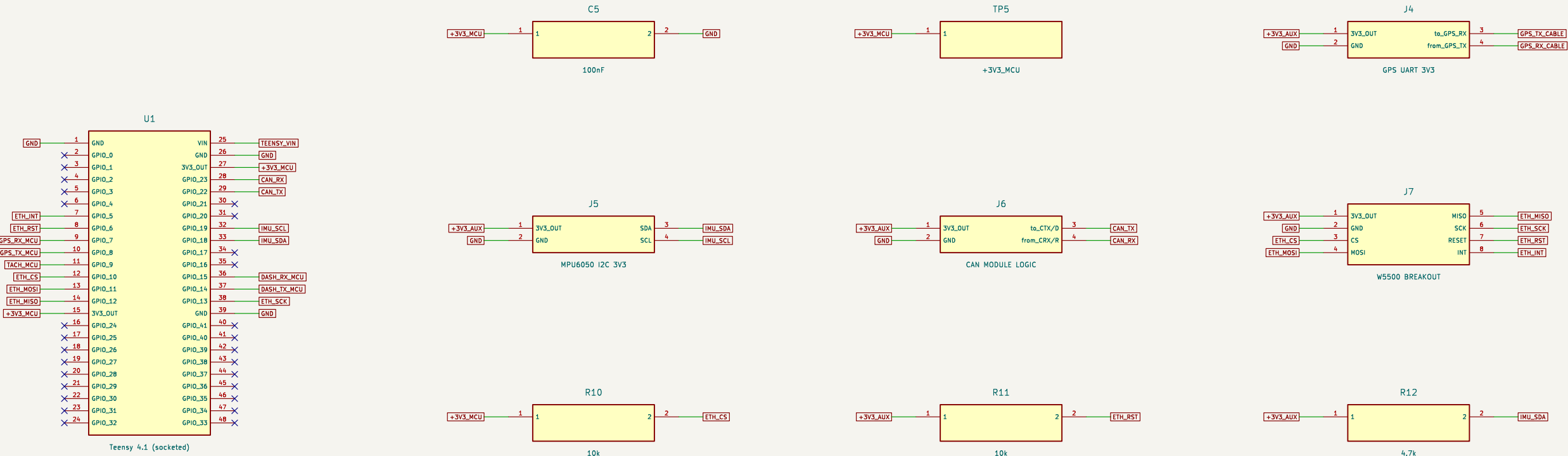
Size: A3

Date:

Rev:

KiCad E.D.A. 9.0.8

Id: 2/5



U1 pad numbering: LEFT 1..24 and RIGHT 25..48, top to bottom viewed from above; USB at top.

External modules not integrated. J6 carries LOGIC, never attach CANH/CANL here. J7 is a custom harness.

No direct oil/coolant ADC conditioning in Rev A. Optional modules must be 3.3V logic.

Do not independently USB-power GPS while attached; it can backfeed its supply or UART.

Sheet: /Teensy 4.1 / expansion connectors/
File: 02-mcu.kicad_sch

Title:

Size: A3

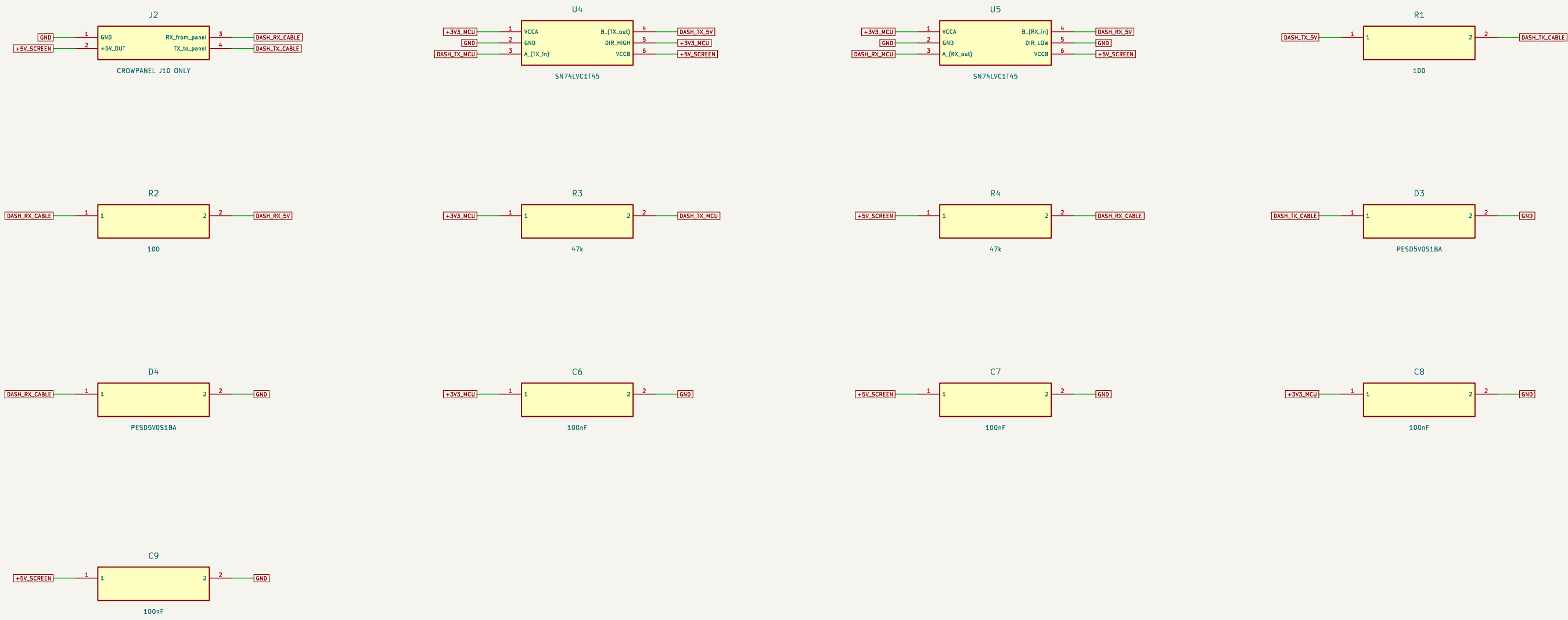
Date:

Rev:

KiCad E.D.A. 9.0.8

Id: 3/5

CrowPanel Advance J10: 5V power and translated UART



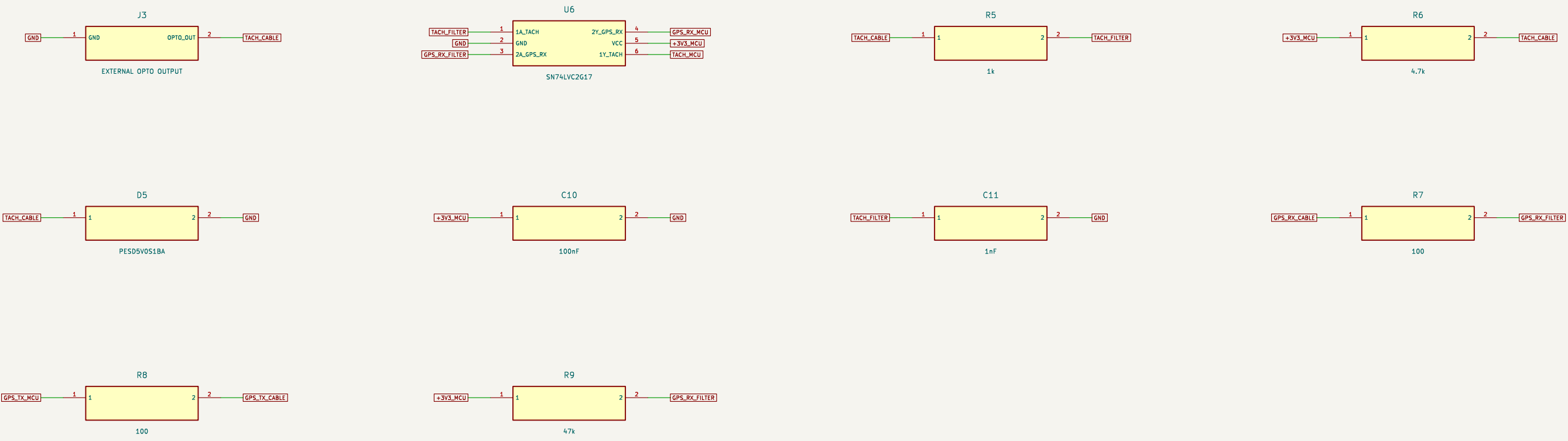
J2 pin3 receives screen TXD0_H; J2 pin4 drives screen RXD0_H. Direction is relative to THIS PCB.

Use the screen ADVANCE J10 5V input header. NEVER connect +5V to screen J2/HY2.0 3V3_OUT.

U4 DIR=HIGH: 3V3 A -> 5V B. U5 DIR=LOW: 5V B -> 3V3 A. Ioff/VCC isolation on translators.

UART TTL. NOT RS232/RS485. Long trunk-to-cabin cable at 921600 baud is NOT yet validated.

External optocoupler and GPS input buffers



J3 accepts external optocoupler transistor output or 0–3.3V/5V logic ONLY; common GND required.

Never raw coil / primary / 12V tach. R6 pulls up to MCU 3V3; R5/C11 low-pass then Schmitt buffer.

U6 channel 2 buffers GPS RX. GPS TX series resistor does not make a 5V or RS232 interface safe.

Pin functions are explicit; block symbols use passive ERC pins. ERC is NOT an electrical safety sign-off.

Sheet: /External optocoupler and GPS input buffers/ File: 04-inputs.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.8		Id: 5/5